

Reza Kermanian

AI & Backend Engineer · Full-Stack Developer · Researcher

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PROFESSIONAL SUMMARY

Multi-disciplinary engineer focused on AI, scalable systems, and global impact. Deep expertise in Python, Django, FastAPI, React, and Next.js. Specialized in AI-driven, data-centric applications using ML, Deep Learning, NLP, and LLMs. Combines software engineering with research writing, trading technologies, and financial market experience. Currently building **Conformis** — an automated EU AI Act compliance platform for European AI startups.

PROFESSIONAL EXPERIENCE

Freelance AI & Backend Developer

2022 — Present

Self-Employed · Remote · International & Local Clients

- Designed and developed scalable backend systems using Python, Django, and FastAPI for production-grade applications.
- Built AI-powered platforms integrating Machine Learning, NLP, and LLM technologies for real-world deployment.
- Developed RESTful APIs and WebSocket integrations with robust authentication and third-party service connectors.
- Implemented data pipelines for structured/unstructured data; architected bilingual React platforms with full RTL/LTR support.
- Focused throughout on clean code, performance optimization, and production readiness.

SELECTED PROJECTS

■ Conformis — EU AI Act Compliance Platform

2025 — Present

Founder & Full-Stack Engineer · conformis.tech

- Built automated EU AI Act compliance scanner from scratch using FastAPI, Next.js, and Claude AI.
- Developed static + AI-powered codebase analyzer detecting risk tiers (prohibited / high / limited / minimal) across Python and TypeScript codebases.
- Integrated GitHub OAuth for direct repo scanning, WeasyPrint for Article 11 PDF generation, and Clerk for auth.
- Deployed on Railway (backend) + Vercel (frontend) with PostgreSQL, PostHog analytics, and Resend email.
- Scanned real open-source projects: LangChain (score 78), CrewAI (score 72).

[FastAPI](#) · [Next.js](#) · [TypeScript](#) · [Claude AI](#) · [PostgreSQL](#) · [Docker](#) · [Railway](#) · [Vercel](#)

Reniom.com — Multi-purpose E-commerce Platform

- Large-scale e-commerce platform featuring authentication, cart system, structured backend logic, and scalable architecture for high-volume traffic.

[React.js](#) · [Django REST](#) · [Authentication](#) · [Scalable Backend](#)

Unimanager.ir — Academic Migration Platform

- Bilingual platform with consultation booking, document submission, user management, and structured student workflow automation for Turkey & Europe applicants.

[React](#) · [Django](#) · [i18next](#) · [RTL/LTR](#) · [Booking System](#)

Chat with PDF — Local Retrieval-Augmented Generation

- Fully offline RAG system for intelligent PDF querying with document chunking, vector embeddings, semantic retrieval, and local LLM integration for privacy-first applications.

Python · FastAPI · Ollama · Sentence Transformers · Vector Search

AI-Powered MetaTrader Trading Robot

- Intelligent trading robot combining CNN, LSTM, and GAN architectures with technical indicators and AI-assisted forecasting for XAUUSD financial market analysis.

CNN · LSTM · GAN · XAUUSD · Algorithmic Trading

AI-Based Intrusion Detection System

- Research-driven IDS leveraging Explainable AI, deep learning, and interpretable security models for next-generation cybersecurity.

XAI · Deep Learning · Cybersecurity · Research

RESEARCH & PUBLICATIONS

AI-Driven Compiler Optimization: A Survey from Rule-Based Heuristics to LLMs2025

Comprehensive review of AI-powered compiler optimization techniques spanning classical heuristic methods to large language model-based approaches.

Systematic Literature Review on Explainable AI Methods for Intrusion Detection2025

Structured review of XAI techniques applied to IDS with comparative analysis of interpretability methods and benchmarking.

Resource Allocation and Task Scheduling in Fog/Edge Computing Using Deep Learning2024

Optimized deep-learning-based model for intelligent resource management and task scheduling in distributed Fog/Edge environments.

TECHNICAL EXPERTISE

Languages

Python · JavaScript · TypeScript · PHP · SQL

Backend

Django · FastAPI · Node.js · Laravel · REST APIs · WebSockets

Frontend

React.js · Next.js · Vite · Tailwind CSS · Framer Motion

AI / ML

LLMs · RAG · NLP · Deep Learning · PyTorch · Hugging Face · Ollama · OpenAI

Data

PostgreSQL · MySQL · Vector DBs · Pandas · NumPy · Sentence Transformers

DevOps

Docker · Git · Linux · CI/CD · EU AI Act · Risk Classification · Audit Docs

Research

Explainable AI · AI Security · Compiler Optimization · Fog/Edge Computing · IDS

EDUCATION & LANGUAGES

B.Sc. Computer Science / Software Engineering

Islamic Azad University (IAU) · Completed
Languages

Persian: Native
Intermediate English